

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. Examination (EE)

November – December 2011

EE 207 NETWORK ANALYSIS

Date: 29.11.2011, Tuesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) The waveform shown in Fig.1(a) is a trapezoidal pulse. Determine the transform $V(s)$. [03]

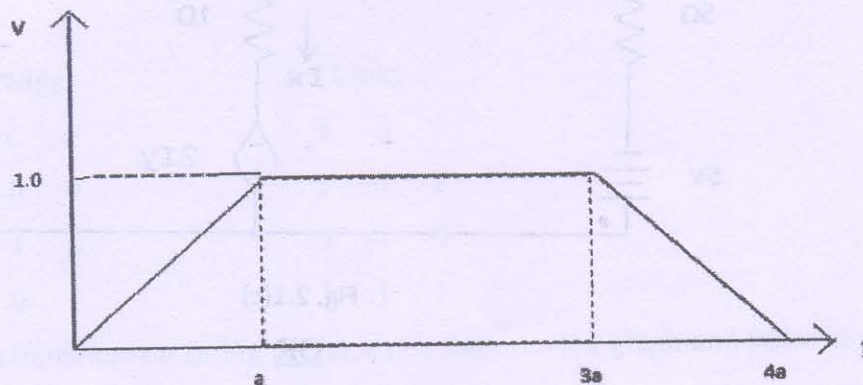


Fig. 1(a)

(b) For the network of Fig. 1(b) the capacitor is initially charged to a voltage V_0 , with a polarity indicated on the diagram. The switch k is closed at time $t = 0$. Determine the particular solution for the current in the circuit where $C = 0.5F$. [04]

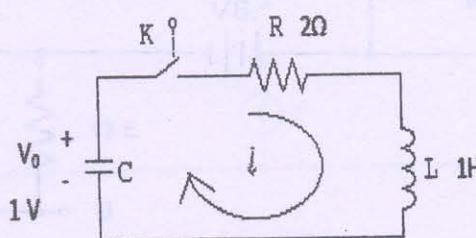


Fig. 1(b)

Q - 2 (a) Define : Node, Branch, Loop and Mesh [02]

(b) Find the power delivered by the 50 V source in the circuit for Fig. 2.1(b). [04]

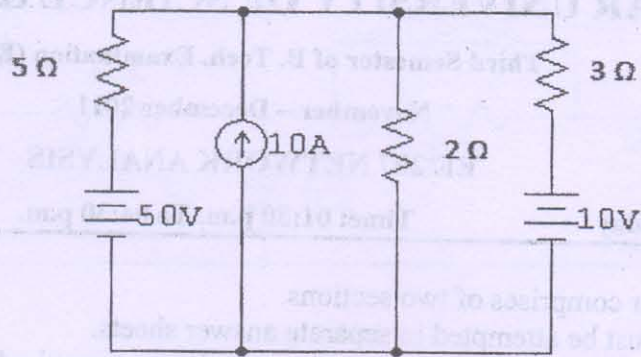


Fig. 2.1(b)

- (c) Find currents I_x and I_y using Mesh analysis for the given network for Fig.2.1(c) [08]

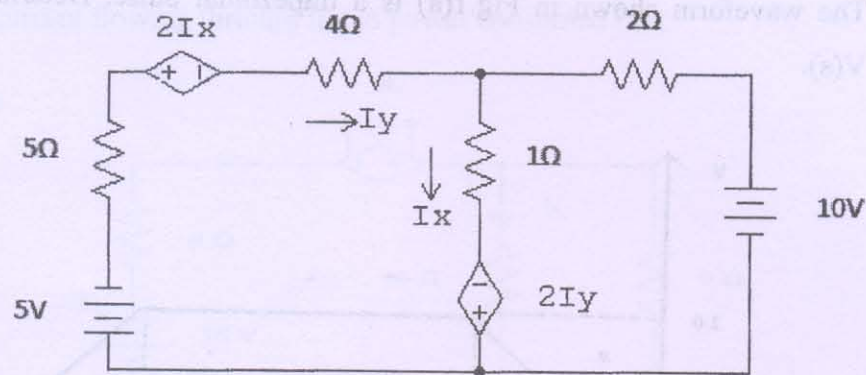


Fig. 2.1(c)

OR

- Q - 2 (a) State Kirchoff's Current Law(KCL) and Voltage Law(KVL). [02]

- (b) Find the voltage of the point A w.r.t. B for Fig. 2.2(a). [04]

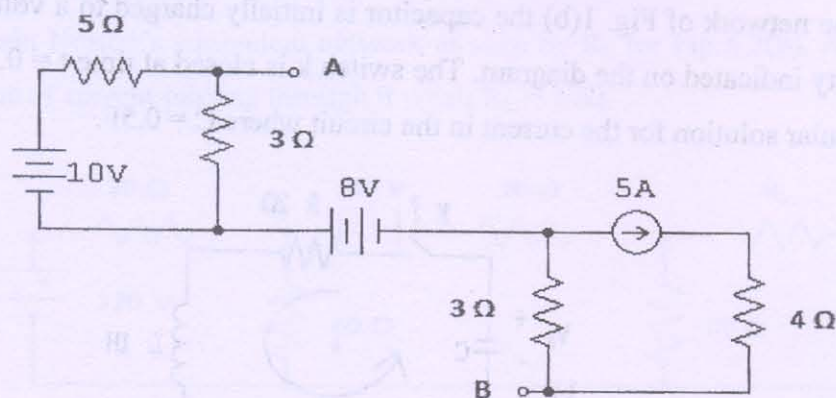


Fig. 2.2(a)

- (c) Find currents I_1 , I_2 , and I_3 and voltages V_a and V_b using nodal analysis for the given network in Fig.2.2(c) [08]

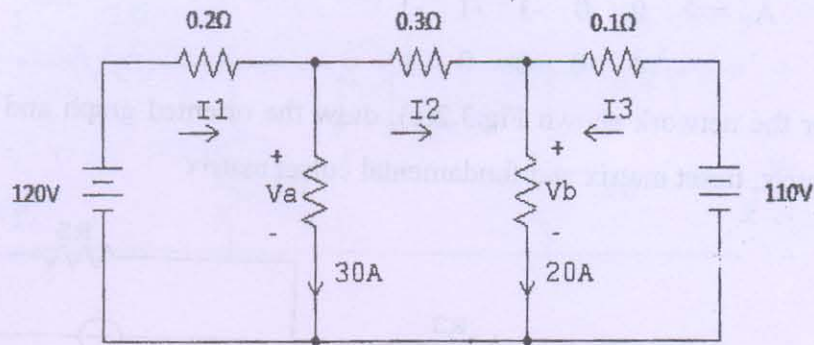


Fig. 2.2(c)

Q - 3 (a) Explain Planar graph, Non-planar graph, Sub-graph and Connected graph in context of Graph Theory. [04]

(b) The fundamental cutset matrix of a network is given as follows. Draw the oriented graph. [04]

Twigs			Links		
a	c	e	b	d	f
1	0	0	1	0	1
0	1	0	0	-1	-1
0	0	1	1	1	1

(c) For the circuit shown in Fig 3.1(c), draw the oriented graph and write the incidence matrix, fundamental cutset matrix and tieset matrix. [06]

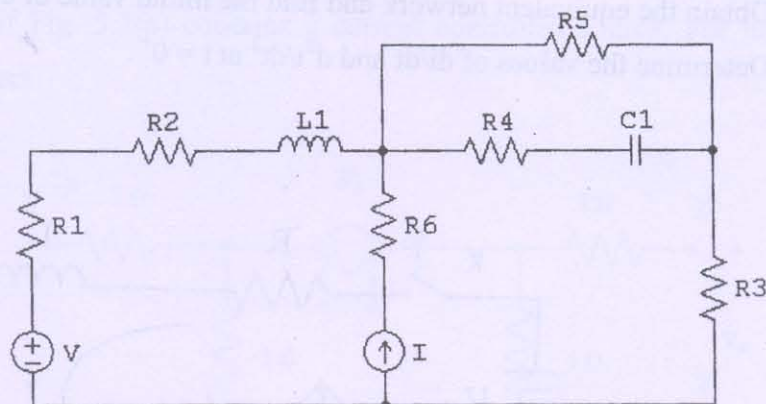


Fig. 3.1(c)

OR

Q - 3 (a) Explain Properties of Rank, Loop, Tree and Co-tree for a graph. [04]

(b) The reduced incidence matrix of an oriented graph is given below, Draw the oriented graph. [04]

$$A = \begin{bmatrix} 0 & -1 & 1 & 1 & 0 \\ 0 & 0 & -1 & -1 & -1 \\ -1 & 0 & 0 & 0 & 1 \end{bmatrix}$$

- (c) For the network shown Fig.3.2(c), draw the oriented graph and write the incidence matrix, tieset matrix and fundamental cutset matrix [06]

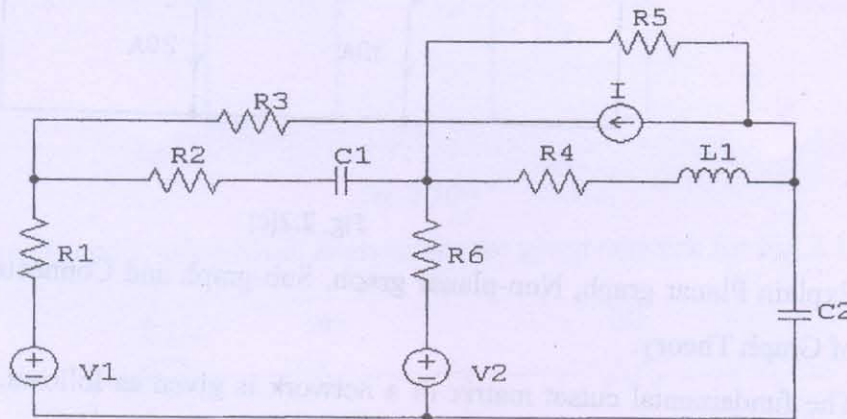


Fig. 3.2(c)

SECTION – II

Q - 4 In the network of the Fig. 4 the switch k is closed at $t = 0$ with the capacitor uncharged and with zero current in the inductor. If $V = 100V$, $R = 10\Omega$, $L = 1H$ and $C = 10\mu F$.

- (a) Obtain the equivalent network and find the initial value of current I at $t = 0^+$. [03]
 (b) Determine the values of di/dt and d^2i/dt^2 at $t = 0^+$. [04]

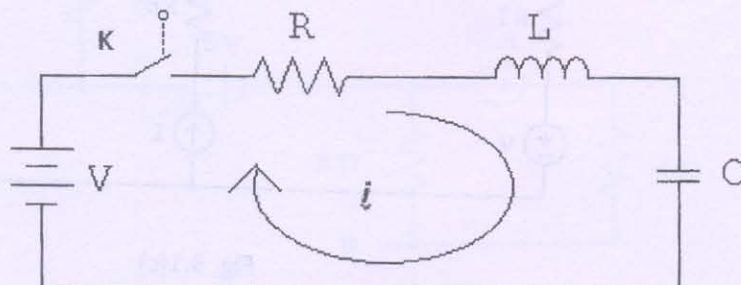


Fig. 4

Q - 5 (a) Find z- parameters for the network of Fig. 5.1(a) [07]

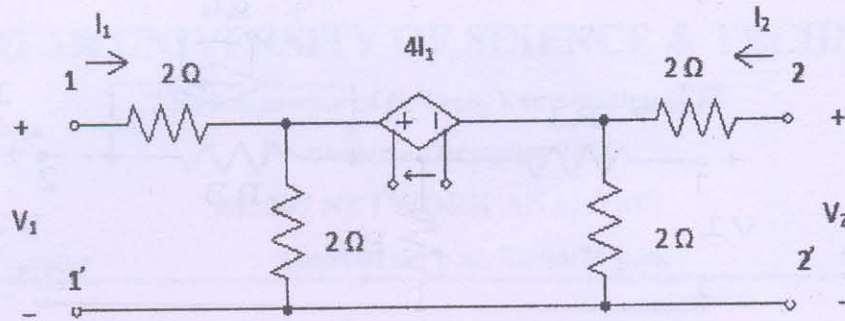


Fig. 5.1(a)

- (b) Find transmission parameters for the two port network shown in Fig. 5.1(b), What [07]
must be the input voltage ' V_1 ' applied for the output voltage V_2 to be to 10V across
the load of 10Ω connected at the port 2.

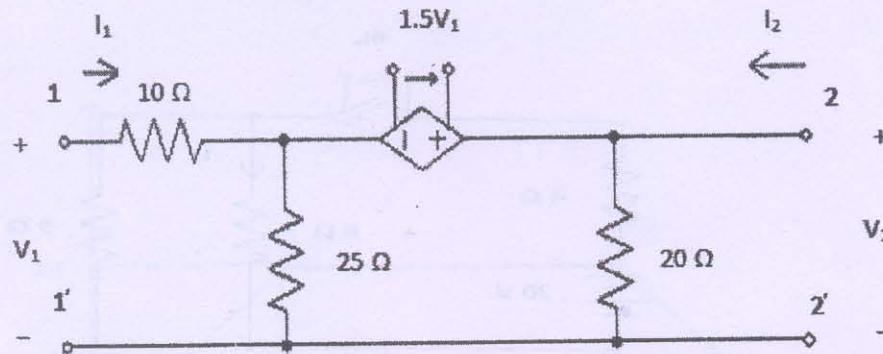


Fig. 5.1(b)

OR

- Q - 5 (a) The network of Fig. 5.2(a) contains a current controlled source. For this network, [07]
find y parameters.

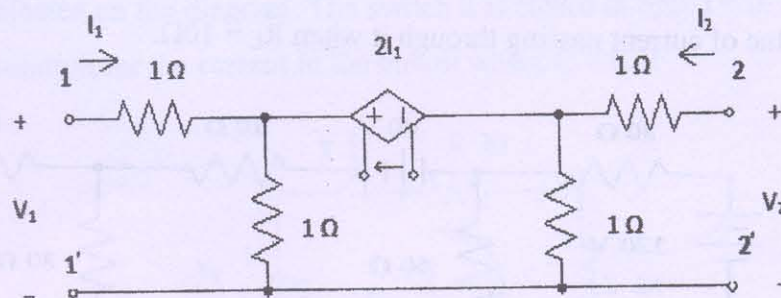


Fig. 5.2(a)

- (b) The network of Fig. 5.2(b) represents a certain transistor over a given range of [07]
frequencies. For this network, Determine h- parameters.

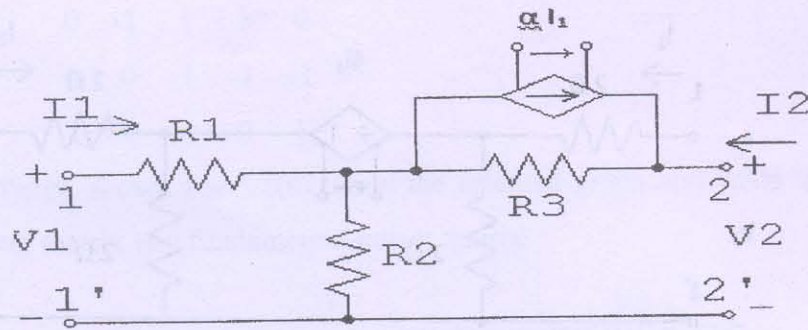


Fig. 5.2(b)

- Q - 6 (a) State Superposition Theorem and mention steps to be followed in superposition theorem. [06]
- (b) Obtain the Thevenin's equivalent network for the 9Ω resistor in the Fig. 6.1(b). Also [08]
find the current flowing through it and power consumed in it.

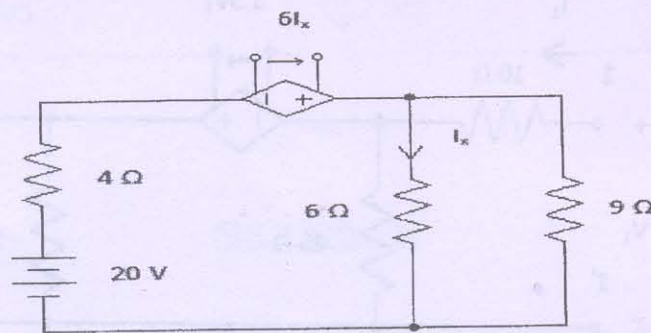


Fig. 6.1(b)

OR

- Q - 6 (a) Determine the Value of R_L for maximum power to be transferred from source to load [06]
for a d.c. network.
- (b) Obtain Norton's equivalent network as seen by R_L for Fig 6.2(b). Also calculate the [08]
value of current passing through it when $R_L = 10\Omega$.

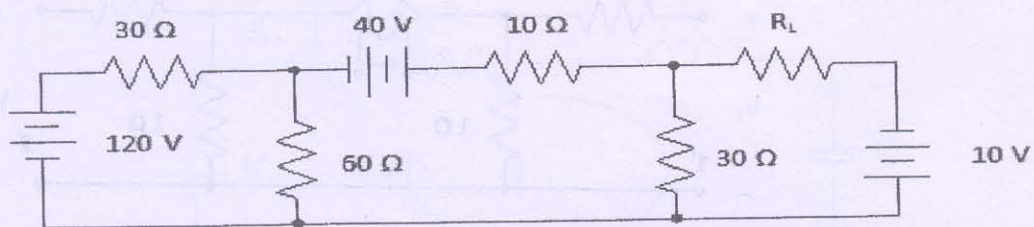


Fig. 6.2(b)
